



## Chapter 3

# How to use HTML to structure a web page

# Objectives

## Applied

1. Code a properly structured HTML web page using the HTML5 semantic elements and any of the other elements that are presented in this chapter.
2. Given the HTML for a web page, code a relative URL that refers to any file in the folder structure for the website.

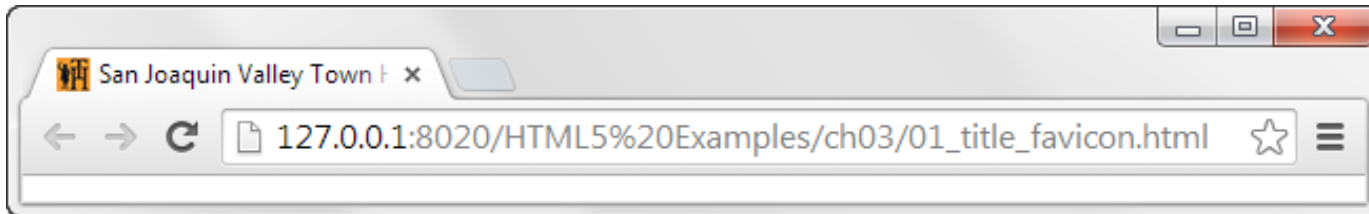
## Knowledge

1. Describe the use of the title and meta elements in the head section of an HTML document.
2. Distinguish between a block element and an inline element.
3. Describe the use of these block elements: h1, h2, h3, and p.
4. Describe the use of these inline elements: br, i, b, sup, em, q, and strong.
5. Describe the use of character entities like &nbsp; or &copy;.

## Objectives (cont.)

6. Describe the use of these core attributes: id, class, and title.
7. Describe the use of the div and span elements in the HTML for a modern website.
8. Describe the use of these HTML5 semantic elements: header, main, section, article, nav, aside, footer, and figure.
9. Distinguish between absolute and relative URLs, and distinguish between root-relative and document-relative paths
10. Distinguish between the use of the <a> element and the img element.
11. Describe the accessibility guidelines for <a> elements and img elements.
12. Describe the two types of lists that you can create with HTML.

## A browser that shows the title and favicon



## A head section with a title and a links to a favicon

```
<head>
  <title>San Joaquin Valley Town Hall |
    speakers and luncheons
</title>
  <link rel="shortcut icon" href="favicon.ico">
</head>
```

## SEO guidelines for the title tag

- Always code a title tag in the head section.
- The title should accurately describe the page's content, and it should include the one or two keywords that you want the page ranked for.
- The title should be interesting enough to entice the reader to click on it when it's shown in the search results for a search engine.
- The title should be unique for each page in your website.
- Limit the length of your titles to around 65 characters because most search engines don't display more than that in their results.

## Internet Explorer note

- A favicon isn't displayed in older versions of Internet Explorer if the page is served from your local file system. It is only displayed if the page is served from the web server.

# A head section that includes metadata

```
<head>
  <title>San Joaquin Valley Town Hall |
    speakers and luncheons
</title>
<meta charset="utf-8">
<meta name="description" content="A yearly lecture
  series with speakers that present new
  information on a wide range of subjects">
<meta name="keywords" content="san joaquin,
  town hall, speakers, lectures, luncheons">
</head>
```

## Three attributes of the <meta> tag

- `charset`
- `name`
- `content`

## SEO guidelines for metadata

- Code the description metadata for each page of your website. It should summarize the contents of the page, it should be unique for each page, and it can be longer than the title tag. When it is displayed in the search-engine results, it should encourage users to click on your link.
- Code the keywords metadata for each page of your website. It should consist of no more than 10 keywords or phrases, and it should be unique for each page.

# block elements for headings and paragraphs

- `h1`
- `h2`
- `h3`
- `h4`
- `h5`
- `h6`
- `p`

# HTML that uses the block elements

```
<h1>San Joaquin Valley Town Hall Programs</h1>
```

```
<h2>Pre-lecture coffee at the Saroyan</h2>
```

```
<p>Join us for a complimentary coffee hour, 9:15 to 10:15  
a.m. on the day of each lecture. The speakers usually  
attend this very special event.</p>
```

```
<h2>Post-lecture luncheon at the Saroyan</h2>
```

```
<p>Extend the excitement of Town Hall by purchasing  
tickets to the luncheons</p>
```

# The block elements in a web browser

## **San Joaquin Valley Town Hall Programs**

### **Pre-lecture coffee at the Saroyan**

Join us for a complimentary coffee hour, 9:15 to 10:15 a.m. on the day of each lecture. The speakers usually attend this very special event.

### **Post-lecture luncheon at the Saroyan**

Extend the excitement of Town Hall by purchasing tickets to the luncheons.

# SEO guidelines for headings and paragraphs

- Use the heading tags to show the structure and importance of the content on a page. Always use the h1 tag to identify the most important information on the page, and only code a single h1 tag on each page. Then, decrease one level at a time to show subsequent levels of importance.
- Don't use heading levels as a way to size text. Instead, use CSS to size the headings.

## Block elements for special types of text

- `pre`
- `blockquote`
- `address`

## HTML that uses the block elements

`<p>How to use JavaScript to display the year:</p>`

`<pre>`

```
var today = new Date();  
document.writeln( today.getFullYear() );
```

`</pre>`

`<p>Ernest Hemingway wrote:</p>`

`<blockquote>Cowardice, as distinguished from panic, is  
almost always simply a lack of ability to suspend the  
functioning of the imagination.  
</blockquote>`

`<p>How to contact Mike Murach & Associates:</p>`

`<address>1-800-221-5528<br>`

```
<a href="mailto:murachbooks@murach.com">  
murachbooks@murach.com</a>
```

`</address>`

# The block elements in a web browser

How to use JavaScript to display the year:

```
var today = new Date();  
document.writeln( today.getFullYear() );
```

Ernest Hemingway wrote:

Cowardice, as distinguished from panic, is almost always simply a lack of ability to suspend the functioning of the imagination.

How to contact Mike Murach & Associates:

*1-800-221-5528*

[murachbooks@murach.com](mailto:murachbooks@murach.com)

## Inline elements for formatting text

- `i`
- `b`
- `sub`
- `sup`
- `br`

## Inline elements for identifying content

- `abbr`
- `cite`
- `code`
- `dfn`
- `em`
- `kbd`
- `q`
- `samp`
- `strong`
- `var`

## HTML that uses some of the inline elements

`<p>If you don't get 78% or more on your final,  
 <em>you won't pass.</em></p>`

`<p>Save a bundle at our <strong>big yearend  
 sale</strong>.</p>`

`<p>When the dialog box is displayed, enter  
 <kbd>brock21</kbd>.</p>`

`<p>The chemical symbol for water is H<sub>2</sub>O.</p>`

`<p><q>To sleep, perchance to dream-ay, there's the  
 rub.</q></p>`

## The inline elements in a web browser

If you don't get 78% or more on your final, *you won't pass.*

Save a bundle at our **big yearend sale.**

When the dialog box is displayed, enter `brock21`.

The chemical symbol for water is  $\text{H}_2\text{O}$ .

"To sleep, perchance to dream-ay, there's the rub."

## Common HTML character entities

| Entity                    | Character                              |
|---------------------------|----------------------------------------|
| <code>&amp;amp;</code>    | &                                      |
| <code>&amp;lt;</code>     | <                                      |
| <code>&amp;gt;</code>     | >                                      |
| <code>&amp;copy;</code>   | ©                                      |
| <code>&amp;reg;</code>    | ®                                      |
| <code>&amp;trade;</code>  | ™                                      |
| <code>&amp;cent;</code>   | ¢                                      |
| <code>&amp;deg;</code>    | °                                      |
| <code>&amp;plusmn;</code> | ±                                      |
| <code>&amp;lsquo;</code>  | ‘ (opening single quote)               |
| <code>&amp;rsquo;</code>  | ’ (closing single quote or apostrophe) |
| <code>&amp;ldquo;</code>  | “ (opening double quote)               |

## Common HTML character entities (cont.)

| Entity                   | Character                |
|--------------------------|--------------------------|
| <code>&amp;rdquo;</code> | ” (closing double quote) |
| <code>&amp;nbsp;</code>  | non-breaking space       |

\_\_\_\_\_

<p>It&rsquo;s time to start your Christmas shopping!</p>

President John F. Kennedy said, "And so, my fellow Americans, ask not what your country can do for you; ask what you can do for your country."

Turning fear into hope, medical futurist Dr. Alan J. Russell will discuss the science of regenerating damaged or diseased human body parts, while offering real hope for the future of human health.

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# The character entities in a web browser

It's time to start your Christmas shopping!

President John F. Kennedy said, "And so, my fellow Americans, ask not what your country can do for you; ask what you can do for your country."

Turning fear into hope, medical futurist Dr. Alan J. Russell will discuss the science of regenerating damaged or diseased human body parts, while offering real hope for the future of human health.

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# Core HTML attributes

- `id`
- `class`
- `title`
- `lang`

# HTML that uses these attributes

```
<html lang="en">
<body>
  <h1>San Joaquin Valley Town Hall</h1>
  <p class="first">Welcome to San Joaquin Valley Town
    Hall.</p>
  <form action="subscribe.php" method="post">
    <p>Please enter your e-mail address to subscribe
      to our newsletter.</p>
    <p class="first field">E-Mail:
      <input type="text" name="email" id="email"
        title="Enter e-mail here."></p>
    <p><input type="submit" value="Subscribe"></p>
  </form>
</body>
</html>
```

## The HTML in a web browser with a tooltip



**San Joaquin Valley Town Hall**

Welcome to San Joaquin Valley Town Hall.

Please enter your e-mail address to subscribe to our newsletter.

E-Mail:

## Accessibility guideline

- Always code the lang attribute on the html element to identify the language for the page.

# The primary HTML5 semantic elements

- `header`
- `main`
- `section`
- `article`
- `nav`
- `aside`
- `footer`

## Accessibility and SEO guideline

- Use the HTML5 semantic elements to indicate the structure of your pages.

# A page structured with header, main, and footer elements

```
<body>
  <header>
    <h1>San Joaquin Valley Town Hall</h1>
  </header>
  <main>
    <p>Welcome to San Joaquin Valley Town Hall. We
      have some fascinating speakers for you this
      season!</p>
  </main>
  <footer>
    <p>&copy; Copyright 2015 San Joaquin Valley Town
      Hall.</p>
  </footer>
</body>
```

## The page displayed in a web browser

# San Joaquin Valley Town Hall

Welcome to San Joaquin Valley Town Hall. We have some fascinating speakers for you this season!

© Copyright 2015 San Joaquin Valley Town Hall.

## Other HTML5 semantic elements

- `time`
- `figure`
- `figcaption`

# The attributes of the time element

- `datetime`
- `pubdate`

## A time element

```
<p>Next year's conference will be on  
  <time datetime="2015-05-31">May 31st</time>.</p>
```

## The figure and figcaption elements

```
<figure>
  <code>
    var today = new Date();<br>
    document.writeln( today.getFullYear() );<br><br>
  </code>
  <figcaption>
    JavaScript code for getting the year
  </figcaption>
</figure>
```

## The code displayed in a browser

```
var today = new Date();
document.writeln( today.getFullYear() );

JavaScript code for getting the year
```

## A block element for structuring a web page

- `div`

## An inline element for formatting text

- `span`

## Accessibility and SEO guidelines

- Use `div` tags only when the HTML5 semantic elements don't apply.
- Use `span` tags only when the tags for identifying content don't apply.

# A page structured with div and span elements

```
<body>
  <div id="header">
    <h1>San Joaquin Valley Town Hall</h1>
  </div>
  <div id="main">
    <p><span id="welcome">Welcome to San Joaquin
      Valley Town Hall.</span> We have some
      fascinating speakers for you this season!</p>
  </div>
  <div id="footer">
    <p>&copy; Copyright 2015 San Joaquin Valley Town
      Hall.</p>
  </div>
</body>
```

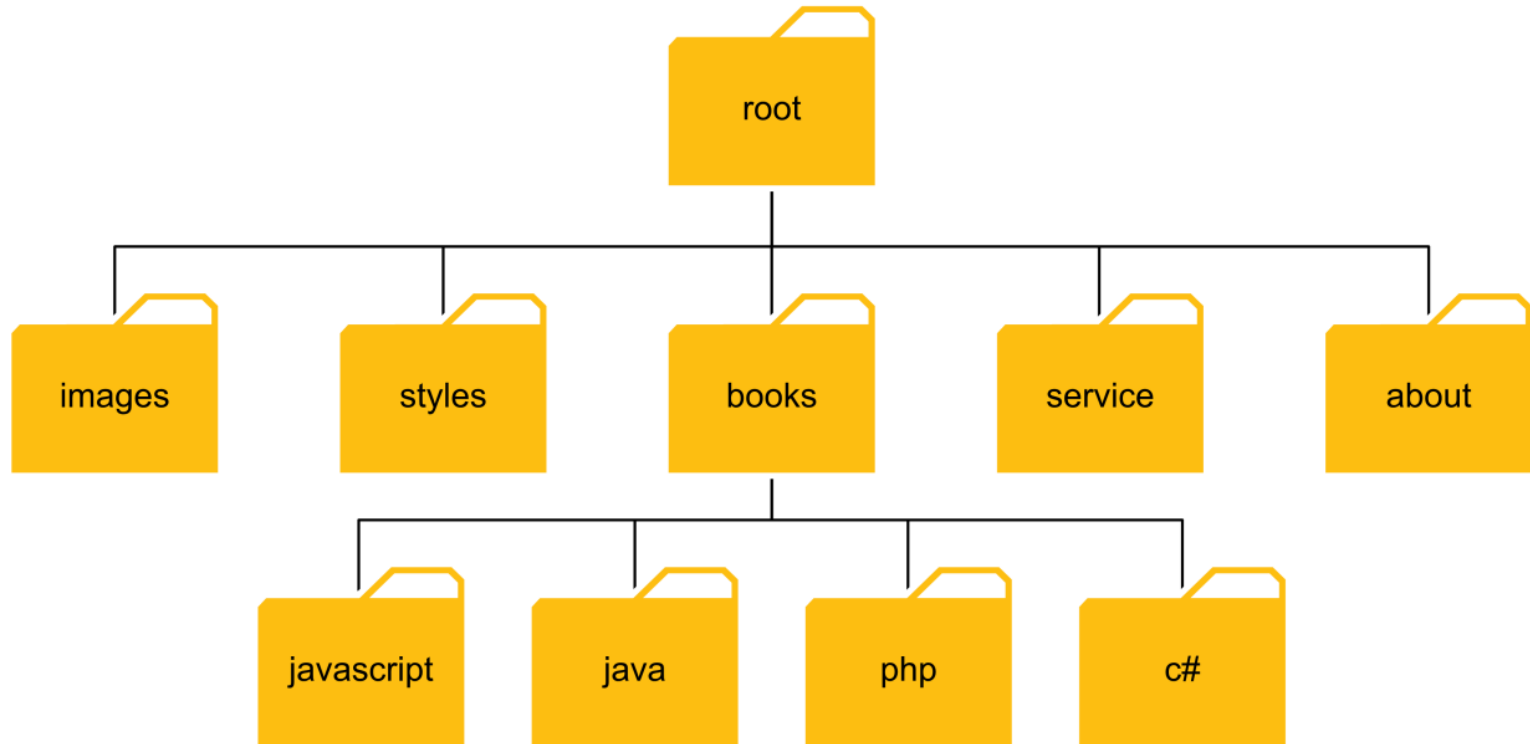
## The page displayed in a web browser

# San Joaquin Valley Town Hall

Welcome to San Joaquin Valley Town Hall. We have some fascinating speakers for you this season!

© Copyright 2015 San Joaquin Valley Town Hall.

# A simple website folder structure



# Absolute URLs

```
http://www.murach.com/index.html  
http://www.murach.com/books/php/toc.html
```

## Root-relative paths

```
/login.html           (refers to root/login.html)  
/images/logo.gif     (refers to root/images/logo.gif)
```

## Document-relative paths that navigate down from the root folder

```
images/logo.gif      (refers to root/images/logo.gif)  
books/php/overview.html (refers to root/books/php/overview.html)
```

## Document-relative paths that navigate up from the root/books folder

```
../index.html         (refers to root/index.html)  
../images/logo.gif   (refers to root/images/logo.gif)
```

## Basic attribute of the <a> element

- href

### A link to a web page in the same folder

```
<p>Go view our <a href="products.html">  
product list</a>.</p>
```

### A link to a web page in a subfolder of the parent folder

```
<p>Read about the <a href="../company/services.html">  
services we provide</a>.</p>
```

### A link to a web page based on the root folder

```
<p>View your <a href="/orders/cart.html">  
shopping cart</a>.</p>
```

### A link to a web page at another website

```
<p>To learn more about JavaScript, visit the  
<a href="http://www.javascript.com/">  
official JavaScript web site</a>.</p>
```

## The links in a web browser

Go view our [product list](#).

Read about the [services we provide](#).

View your [shopping cart](#).

To learn more about JavaScript, visit the [official JavaScript website](#).

## SEO and accessibility guideline

- The content of a link should be text that clearly indicates where the link is going.

## Elements that create ordered and unordered lists

- `<ul>`
- `<ol>`
- `<li>`

## HTML that creates two lists

```
<p>We have books on a variety of languages, including</p>
<ul>
  <li>JavaScript</li>
  <li>PHP and MySQL</li>
  <li>Servlets and JSP</li>
  <li>ASP.NET</li>
</ul>

<p>You will need to complete the following steps:</p>
<ol>
  <li>Enter your billing information.</li>
  <li>Enter your shipping information.</li>
  <li>Confirm your order.</li>
</ol>
```

# The lists in a web browser

We have books on a variety of languages, including

- JavaScript
- PHP
- Java
- C#

You will need to complete the following steps:

1. Enter your billing information.
2. Enter your shipping information.
3. Confirm your order.

## Attributes of the <img> element

- `src`
- `alt`
- `height`
- `width`

## An img element

```
  
<h1>Mike Murach & Associates, Inc.</h1>
```

## The image in a web browser



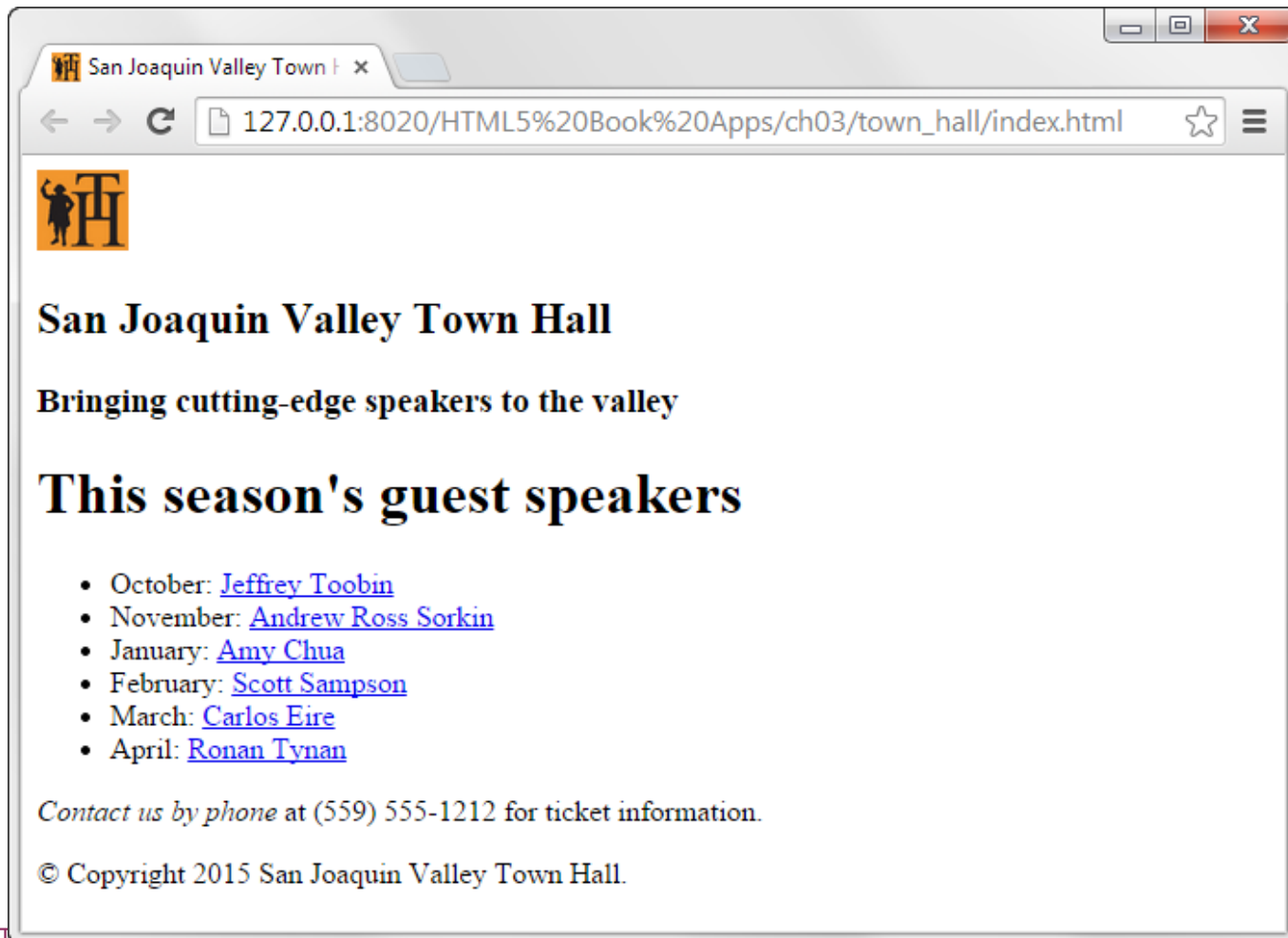
## The image formats that are supported by most browsers

- JPEG (Joint Photographic Experts Group)
- GIF (Graphic Interchange Format)
- PNG (Portable Network Graphics)

## Accessibility guidelines

- For images with useful content, always code an alt attribute that describes the image.
- For images that are used for decoration, code the alt attribute with no value ("").

## A web page that uses some of the HTML presented in this chapter



# The HTML file for the web page

```
<!DOCTYPE html>

<html lang="en">
  <head>
    <meta charset="utf-8">
    <title>San Joaquin Valley Town Hall</title>
    <link rel="shortcut icon" href="images/favicon.ico">
  </head>

  <body>
    <header>
      
      <h2>San Joaquin Valley Town Hall</h2>
      <h3>Bringing cutting-edge speakers to the valley</h3>
    </header>

    <main>
      <h1>This season's guest speakers</h1>
      <nav>
        <ul>
          <li>October: <a href="speakers/toobin.html">
            Jeffrey Toobin</a></li>
```

## The HTML file for the web page (cont.)

```
<li>November: <a href="speakers/sorkin.html">
    Andrew Ross Sorkin</a></li>
<li>January: <a href="speakers/chua.html">
    Amy Chua</a></li>
<li>February: <a href="speakers/sampson.html">
    Scott Sampson</a></li>
<li>March: <a href="speakers/eire.html">
    Carlos Eire</a></li>
<li>April: <a href="speakers/tynan.html">
    Ronan Tynan</a></li>
</ul>
</nav>

<p><em>Contact us by phone</em> at (559) 555-1212
    for ticket information.</p>
</main>

<footer>
    <p>&copy; Copyright 2015 San Joaquin Valley Town
        Hall.</p>
</footer>
</body>
</html>
```

## Exercise 3-1 Enter the HTML for the home page



### San Joaquin Valley Town Hall

Celebrating our 75<sup>th</sup> Year

### Our Mission

San Joaquin Valley Town Hall is a non-profit organization that is run by an all-volunteer board of directors. Our mission is to bring nationally and internationally renowned, thought-provoking speakers who inform, educate, and entertain our audience! As one of our members told us:

“Each year I give a ticket package to each of our family members. I think of it as the gift of knowledge...and that is priceless.”

### Our Ticket Packages

- Season Package: \$95
- Patron Package: \$200
- Single Speaker: \$25

### This season's guest speakers

October  
[Jeffrey Toobin](#)



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## Short 3-1 Create an HTML page for a speaker

### Andrew Ross Sorkin, author of *Too Big to Fail*

November 2015



New York Times columnist and author, Andrew Ross Sorkin, has been described as “the most famous financial journalist of his generation.” His book, *Too Big to Fail*, was the first behind-the-scenes account of the financial crisis that led to our current recession.

The Economist, The Financial Times, and Business Week all named *Too Big To Fail* one of the best books of the year. The book was published by Viking October 20, 2009. The book was adapted as a movie by HBO Films and premiered on HBO on May 23, 2011.

#### The cast of the movie: *Too Big to Fail*

1. William Hurt as Hank Paulson
2. Paul Giamatti as Ben Bernanke
3. Billy Crudup as Timothy Geithner

#### Movie reviews

“Lots of heat but not very much light. It came across as a kind of daytime soap opera version showing the personalities of the players rather than dealing with the nuts and bolts of what really happened. Even that would not have been so bad but the movie tries to portray lots of these folks as sympathetic figures just trying to do the right thing.”

“I am watching this for the 3rd time, trying to understand how this large group of capitalist pigs could royally screw their families, descendants, neighbors, countrymen and the rest of the whole world with absolutely no compassion.”

Prepared by: Your Name

Date: Today's Date

## Halloween 3 Create the home page

